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MODEL SELECTION AND VALIDATION REPORT ( haul_duration model )
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## OBSERVATIONS:3449

## SANITY: OK − all models in the selection process passed sanity checks.

## SELECTED RANDOM TERM: spatiotemporal_random_fields_ar1

## MODEL SUMMARY ( haul_duration model )

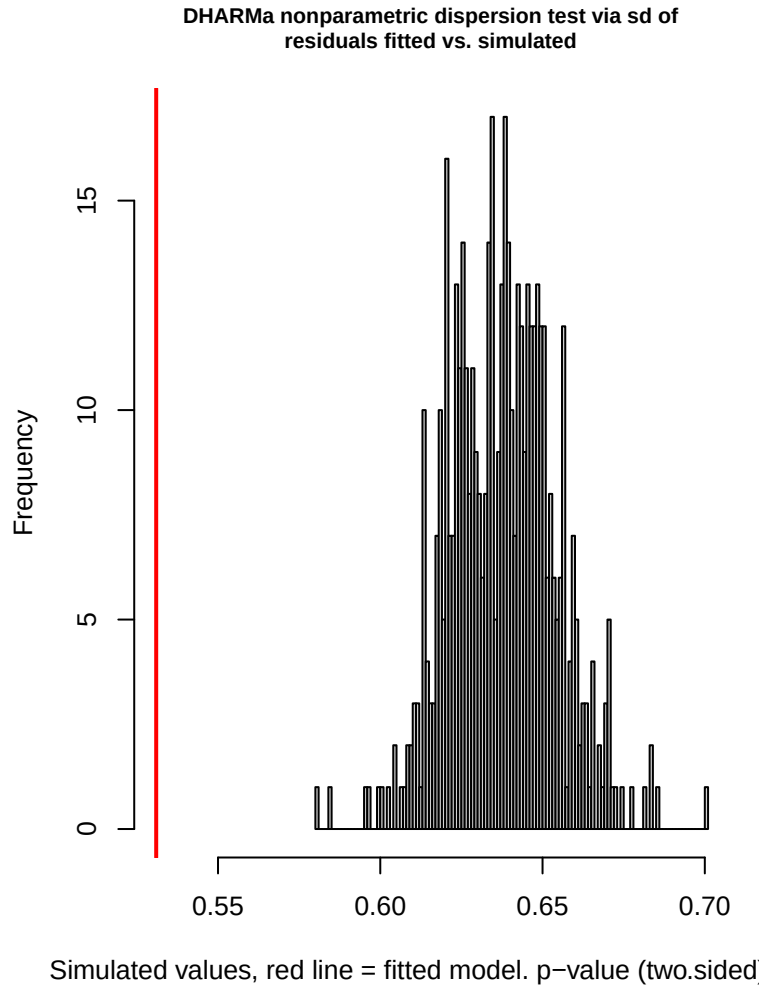
Spatiotemporal model fit by REML ['sdmTMB']
Formula: fixed_formula
Time column: year
Family: gaussian(link = 'identity')

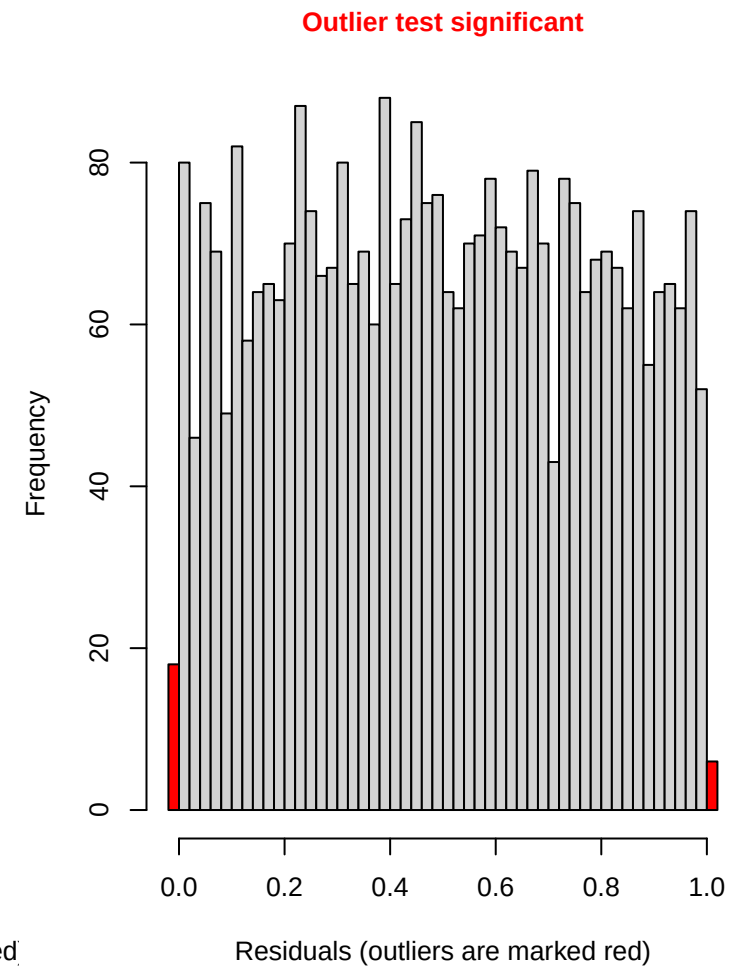
Conditional model:
            coef.est coef.se
(Intercept)    11.06    2.23
haul_dur        0.06    0.03

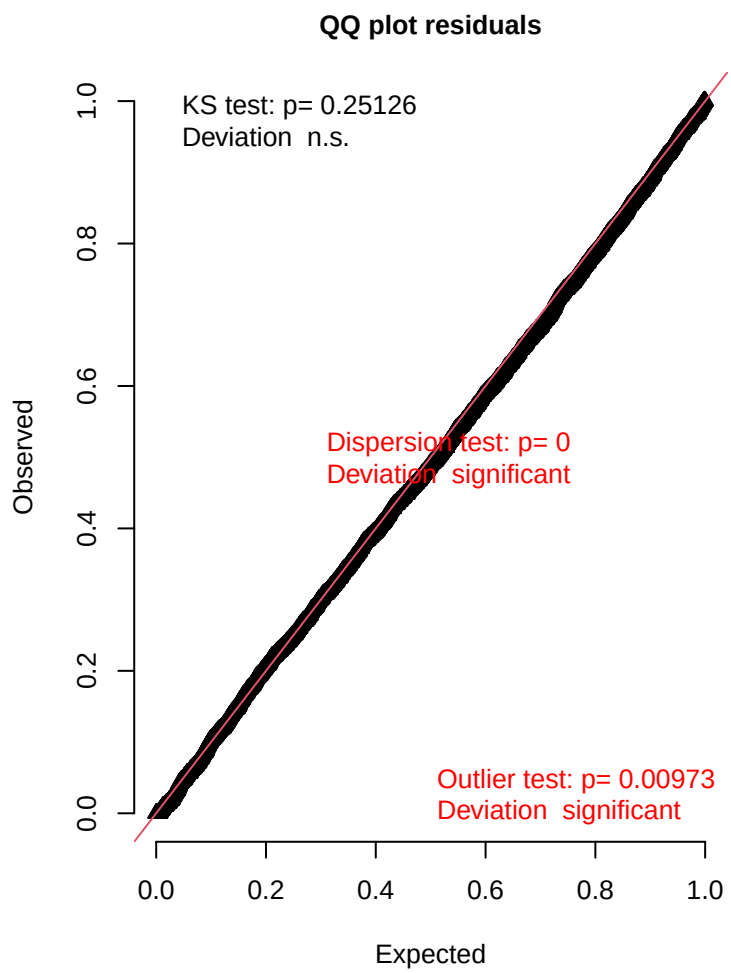
Dispersion parameter: 2.42
Spatiotemporal AR1 correlation (rho): 0.98
Matérn range: 541.80
Spatiotemporal marginal AR1 SD: 4.28
REML criterion at convergence: 8272.110

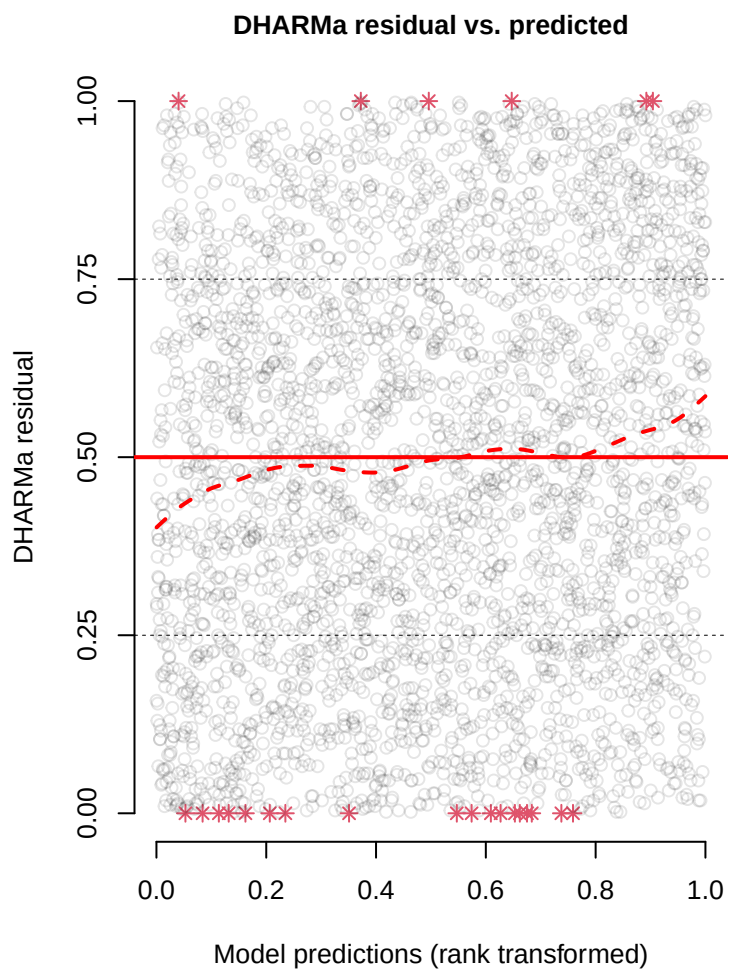
See ?tidy.sdmTMB to extract these values as a data frame.

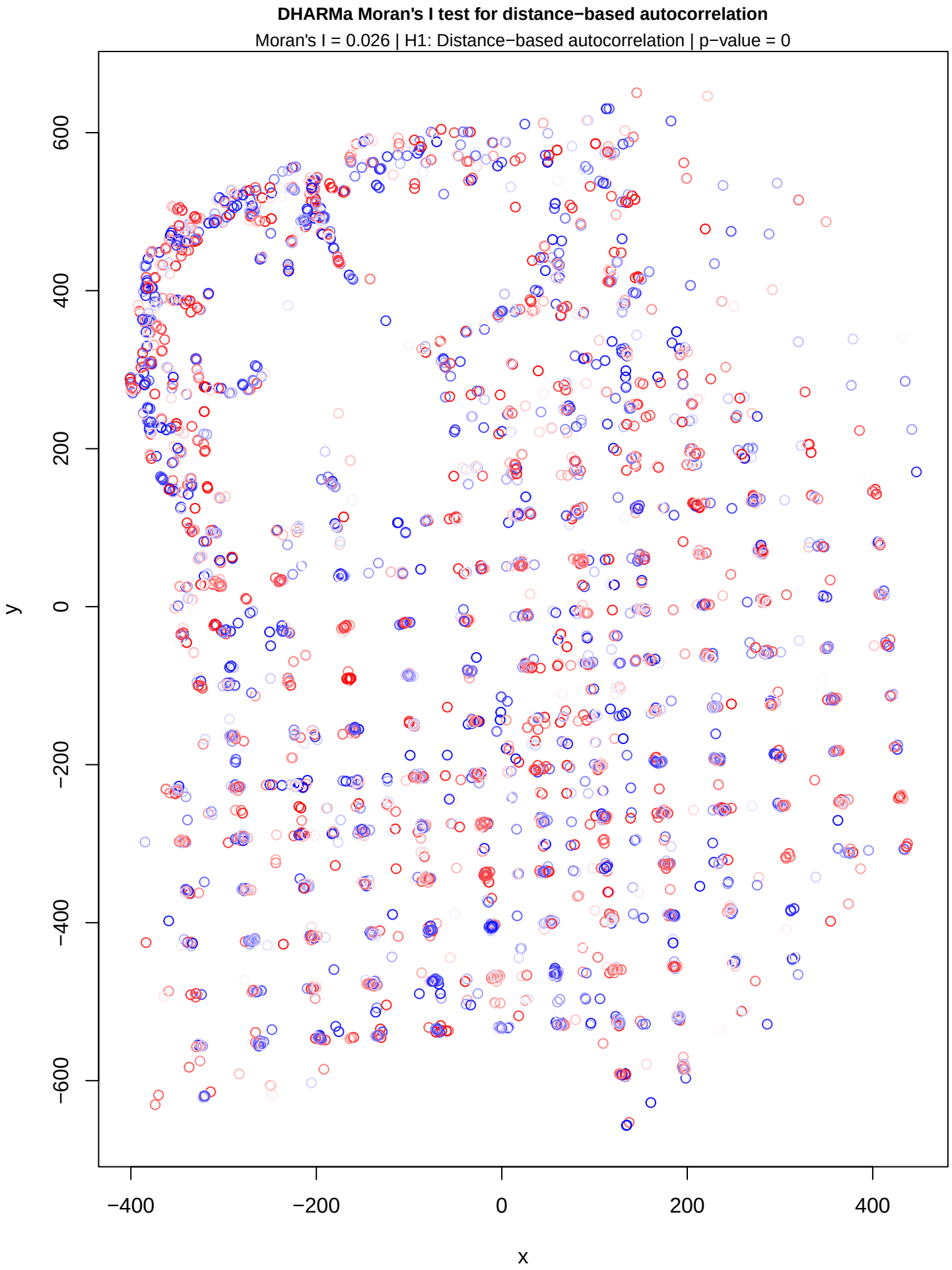
## SELECTION STATISTICS
                                                    df      AIC  delta_AIC
spatiotemporal_random_fields_ar1                     6 16556.22   2.991426
spatiotemporal_random_fields_ar1_with_anisotropy     8 16559.21 458.426664
spatiotemporal_random_fields_iid                     5 17017.64   1.775368
spatiotemporal_random_fields_iid_with_anisotropy     7 17019.41  64.430435
spatial_random_fields                                5 17083.84   3.730073
spatial_random_fields_with_anisotropy                7 17087.57 640.146231
error_only                                           3 17727.72         NA
```

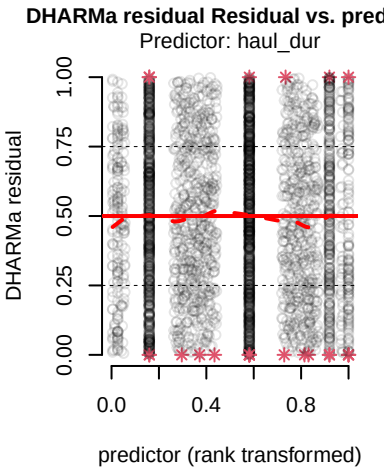

DHARMa nonparametric dispersion test via sd of residuals fitted vs. simulated
Frequency
0
5
10
15
0.55
0.60
0.65
0.70
Simulated values, red line = fitted model. p−value (two.sided


Outlier test significant
Frequency
0
20
40
60
80
0.0
0.2
0.4
0.6
0.8
1.0
Residuals (outliers are marked red)


QQ plot residuals
KS test: p= 0.25126
Deviation n.s.
Dispersion test: p= 0
Deviation significant
Outlier test: p= 0.00973
Deviation significant
Observed
0.0
0.2
0.4
0.6
0.8
1.0
Expected


DHARMa residual vs. predicted
DHARMa residual
0.00
0.25
0.50
0.75
1.00
0.0
0.2
0.4
0.6
0.8
1.0
Model predictions (rank transformed)


DHARMa Moran's I test for distance−based autocorrelation
Moran's I = 0.026 | H1: Distance−based autocorrelation | p−value = 0
y
600
400
200
0
−200
−400
−600
x
−400
−200
0
200
400


DHARMa residual Residual vs. pred
Predictor: haul_dur
DHARMa residual
0.00
0.25
0.50
0.75
1.00
0.0
0.4
0.8
predictor (rank transformed)